STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 3: Motion in a Plane

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. Is displacement scalar or vector? [1]
2. x-component at angle theta? [1]
3. Horizontal acceleration of an ideal projectile? [1]
4. Direction of centripetal acceleration? [1]
5. Magnitude of perpendicular components 3 and 4? [1]
6. Relation between v and omega? [1]
7. Ideal maximum-range angle at equal heights? [1]
8. Vertical velocity at projectile peak? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. Resolve 10 N at 60 deg . [2]
10. Displacement after a full circular lap? [2]
11. v=12 m/s, r=6 m: a_c? [2]
12. Horizontal launch from 20 m, g=10: flight time? [2]
13. Perpendicular vectors 6 and 8: resultant? [2]
14. 120 rpm in rad/s? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. Projectile at 20 m/s, 30 deg , g=10: T,H? [3]
16. Forces 8 N east and 6 N north: resultant? [3]
17. Cyclist at 10 m/s, radius 25 m: a_c? [3]
18. 25 m/s at 37 deg , sin=.6, cos=.8, g=10: time/range? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. Projectile 20 m/s at 30 deg : time, height, range? [4]
20. 5 kg at 4 m/s in radius 2 m: inward force and work? [4]

End of paper - check working, units, signs and significant figures.